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# Generative AI for Zero-Day Attack Detection

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## 1.1 Research Methodology

This study adopts a combined methodological approach consisting of (a) a literature review, (b) the use of Generative AI tools for content generation and analysis, and (c) a comparative evaluation of the obtained results.

Initially, a comprehensive review of academic and technical sources was conducted, focusing on zero-day attack detection and the application of machine learning and Generative AI techniques in cybersecurity.

Subsequently, two Generative AI models (ChatGPT and Claude 3) were employed to produce analytical content on the subject. The generated outputs were systematically evaluated based on their technical accuracy, depth of analysis, and effectiveness in supporting the understanding of the problem.

Finally, a comparative assessment of the two models was carried out to identify their respective strengths and limitations, as well as to examine the role of Generative AI tools as supportive instruments in the analysis of complex cybersecurity challenges.

# PART A: Theoretical Analysis and Current Applications

## 2.1 Introduction to the Concept of Zero-Day

Zero-day attacks represent one of the most critical challenges in modern cybersecurity, as they exploit previously unknown software vulnerabilities for which no patches or mitigation measures are yet available. Their stealthy and unpredictable nature makes timely detection particularly difficult.

A prominent example is *Stuxnet* (2010), widely regarded as the first cyberweapon designed to cause physical damage. Targeting Iran's nuclear program, Stuxnet exploited multiple zero-day vulnerabilities in Windows systems and Siemens industrial control systems, enabling it to manipulate the speed of centrifuges while simultaneously concealing its presence from system operators.

In 2017, the exploitation of *EternalBlue*—an exploit originally developed by the U.S. National Security Agency (NSA)—facilitated the rapid global spread of the *WannaCry* ransomware, leading to widespread operational disruptions across multiple sectors.

More recently, the discovery of the *Log4Shell* vulnerability (2021) in the widely used Log4j library underscored the risks associated with software supply chains. Its exploitation enabled remote code execution across millions of applications worldwide, exposing even critical infrastructures to significant threats.

The detection of such attacks remains particularly challenging due to the absence of known signatures or predefined behavioral patterns. Consequently, there is an increasing need for advanced detection mechanisms, with artificial intelligence emerging as a key enabler in addressing this challenge.

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## 2.2 Generative AI – Basic Principles

The field of Artificial Intelligence (AI) has evolved rapidly over recent decades, focusing on the development of computational systems capable of simulating human cognitive functions such as reasoning, problem-solving, and natural language understanding (Russell & Norvig, 2020). Within this broader context, Generative AI models represent a distinct subfield, as they are not

limited to analyzing or classifying data but are specifically designed to generate new content—including text, images, audio, and even code (Goodfellow et al., 2016).

The development of these models is primarily based on deep learning techniques, with transformer-based architectures forming the foundation of the most advanced systems. Representative examples include OpenAI's GPT-4 (2023), Anthropic's Claude (2024), and Google DeepMind's Gemini (2024). These models are trained on vast amounts of data, enabling them to capture complex patterns in human communication and produce outputs that closely resemble natural language.

More specifically, Generative AI systems model the underlying probability distributions of their training data. Through this process, they become capable of generating new content that, while novel, remains consistent with the patterns they have learned. Their applications extend beyond content generation to areas such as decision support, synthetic data generation for training other models, and, more recently, threat detection in cybersecurity (Moustafa et al., 2023).

Despite their significant potential, the use of Generative AI requires careful and critical evaluation, particularly in domains such as information security, where accuracy, reliability, and robustness are essential.

## 2.3 Zero-Day Detection with AI

The detection and mitigation of zero-day attacks remain among the most challenging problems in modern cybersecurity. Since the vulnerabilities exploited by such attacks are unknown to developers and security professionals, traditional detection methods—such as signature-based approaches relying on known threat databases—are often ineffective.

In this context, Artificial Intelligence (AI), and particularly Machine Learning (ML), emerges as a powerful tool. A key strength of machine learning models lies in their ability to learn the normal behavior of a system or network and subsequently identify deviations that may indicate malicious activity.

In many cases, supervised learning techniques are employed, where models are trained on labeled datasets containing instances classified as "normal" or "malicious." However, this approach faces inherent limitations when dealing with zero-day attacks, as no prior labeled data exist for previously unseen threats.

For this reason, unsupervised learning techniques are often more suitable. Instead of relying on predefined categories, these methods focus on modeling normal system behavior and detecting anomalies that may correspond to novel or unknown threats. Techniques such as anomaly detection and clustering play a central role in this approach.

In addition, reinforcement learning is increasingly being explored in dynamic environments, where systems continuously adapt their detection strategies based on ongoing interactions with networks or organizational infrastructures.

It is also important to highlight the role of modern deep neural networks, particularly transformer-based architectures, which have significantly improved the ability to capture complex patterns in large-scale data. In applications such as network traffic analysis and malware detection, these models have demonstrated strong performance, even when dealing with previously unseen attack patterns.

Finally, Generative AI introduces an additional layer of capability in zero-day defense. Through the generation of synthetic data and the simulation of attack scenarios, these models can enhance system robustness by enabling proactive preparation for threats that have not yet been observed.

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## 2.4 Application Examples & Trends

The application of Generative AI in the detection of zero-day attacks has gained significant momentum in recent years within the cybersecurity domain. One of the most prominent practices is the generation of synthetic data to enhance training datasets. Through the use of Generative Adversarial Networks (GANs), researchers can create artificial network traffic samples or simulated log data, thereby improving the ability of detection systems to identify emerging and previously unseen threats.

At the same time, autoencoder-based models—and particularly Variational Autoencoders (VAEs)—play a critical role in anomaly detection. By learning the normal behavior of a system and identifying deviations during the data reconstruction process, these models provide an effective mechanism for detecting novel or sophisticated attack patterns.

Similarly, Large Language Models (LLMs), such as GPT and BERT, are increasingly being utilized in the analysis of cybersecurity data. When trained on log files and network activity, these models can uncover behavioral patterns that may not be easily identifiable by human analysts and, in some cases, support the early detection of potential attacks.

Furthermore, Generative AI is widely used to simulate realistic cyberattack scenarios. These simulations enable cybersecurity teams to evaluate the resilience of their systems against previously unseen threats, often through Red Team–Blue Team exercises.

Among the emerging trends, the development of AI-driven Extended Detection and Response (XDR) solutions is particularly noteworthy, as they enable unified threat detection and response across endpoints, networks, and cloud environments. Additionally, fine-tuning large language models with domain-specific cybersecurity data has shown considerable potential in improving detection accuracy for complex threats.

Finally, it is increasingly recognized that the adoption of AI-based systems in cybersecurity depends heavily on transparency and trust. As a result, Explainable AI (XAI) and the development of multimodal models—capable of integrating diverse data types such as text, images, and network data—are expected to play a central role in the future evolution of the field.

## 2.5 Limitations and Risks

Despite the significant potential of Generative AI in detecting zero-day attacks, its adoption is accompanied by important limitations and challenges that must be carefully considered.

One major concern relates to the accuracy of these systems. Models may generate false positives by flagging benign activity as malicious or, more critically, fail to detect actual threats. This issue is particularly pronounced when training datasets are incomplete or fail to capture the full diversity of real-world attack scenarios.

Another key limitation is the lack of transparency, often referred to as the “black-box” nature of modern AI systems. In deep learning models, decision-making processes are frequently difficult to interpret, making it challenging for security analysts to understand the rationale behind specific detections or classifications. This lack of explainability can reduce trust and hinder effective adoption.

In addition, Generative AI models typically require substantial computational resources for both training and deployment. The need for high-performance hardware, such as advanced GPUs and large memory capacity, may limit their accessibility, particularly for smaller organizations.

From a technical perspective, overfitting remains a critical issue. When models become overly specialized to their training data, their ability to generalize to new and unseen threats is significantly reduced. This problem is often exacerbated by biases in the training data, which may lead to uneven detection performance across different types of attacks.

Furthermore, Generative AI technologies may themselves be exploited by malicious actors. Their capability to generate realistic text, code, or attack scenarios can be leveraged to develop more sophisticated malware or to conduct advanced social engineering attacks.

Data security and privacy also represent critical concerns. During training or operation, models may inadvertently store or reproduce sensitive information, potentially creating vulnerabilities that could be exploited.

Finally, the susceptibility of AI models to adversarial attacks—where carefully crafted inputs are designed to mislead the system—introduces additional risks. In such cases, detection mechanisms may be bypassed, allowing malicious activity to remain undetected.

Overall, while Generative AI holds significant promise for advancing cyber-threat detection, its effective deployment requires careful design, continuous validation, and the integration of human oversight to ensure reliability and security.

## 2.6 Proposed Architecture for Zero-Day Detection using Generative AI

To illustrate the practical application of Generative AI in zero-day attack detection, a conceptual system architecture is proposed.

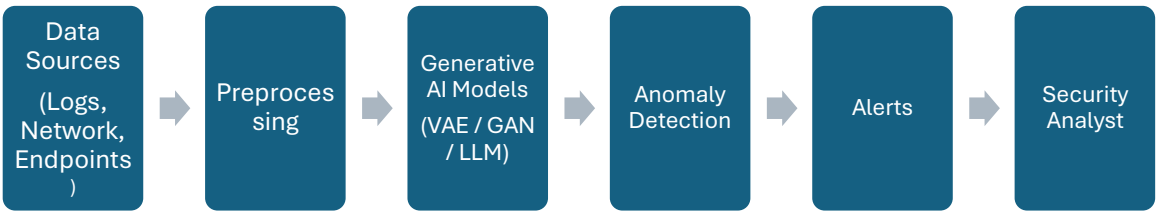
The architecture is based on the collection and integration of data from multiple sources, including log files, network traffic, and endpoint activity. These data are first preprocessed and transformed into a suitable format for analysis by machine learning models.

Subsequently, Generative AI techniques—such as Variational Autoencoders (VAEs) and Generative Adversarial Networks (GANs)—are employed to model normal system behavior and to generate synthetic data that enhance the robustness of the detection process.

In parallel, Large Language Models (LLMs) can be utilized for log analysis and the interpretation of potential threats. By analyzing patterns and contextual information, the system identifies deviations from normal behavior and generates alerts, which are then subject to evaluation by cybersecurity analysts.

Human oversight remains essential for validating detections and supporting decision-making processes. It should be noted that the proposed architecture does not represent a fully implemented system, but rather a conceptual framework intended to demonstrate the role of Generative AI technologies in the detection of advanced cyber threats.

*Figure 1: Proposed Architecture for Zero-Day Detection using Generative AI*




The figure illustrates the data flow from collection and preprocessing to anomaly detection and alert generation, highlighting the contribution of Generative AI models in identifying previously unknown threats.

# PART B: Differential Evaluation of AI

## 3.1 Use of a Second Generative AI Model (Claude 3)

In order to conduct a comparative evaluation within the scope of this study, it was considered necessary to employ a second Generative AI model, distinct from the primary tool used in the initial stages of the analysis. For this purpose, Claude 3, a contemporary artificial intelligence system developed by Anthropic, was selected.

The choice of Claude 3 was based on its strong reputation in the field of artificial intelligence, as well as its capability to generate structured, accurate, and analytically rich content. The model is designed to balance technical depth with clarity, while also demonstrating a more cautious and responsible approach to content generation compared to other existing systems.

The following prompt was provided to the model:

*"Write an analytical academic presentation on the topic: Generative AI for Zero-Day Attack Detection. Describe zero-day attacks, explain the operation of Generative AI, discuss applications of GANs, VAEs, LLMs, and XDR, and analyze the limitations of this approach."*

The response generated by Claude 3 addressed the topic comprehensively, covering multiple key thematic areas. It provided a detailed explanation of zero-day attacks, outlined the core principles of Generative AI systems, and examined practical applications of architectures such as GANs, VAEs, and LLMs in the cybersecurity domain. Furthermore, the model incorporated references to real-world scenarios and highlighted both the advantages and the limitations associated with these technologies.

Notably, the generated content extended beyond purely theoretical discussion, placing particular emphasis on real-world applicability. This practical orientation enhances the usefulness of the model as a tool for analyzing complex cybersecurity challenges, especially in the context of emerging threats.

Although the response demonstrated a high level of technical accuracy, its style was more narrative compared to strictly formal academic writing. As such, Claude 3 can be considered a valuable tool for producing comprehensive and accessible analyses, particularly in interdisciplinary and rapidly evolving areas such as the application of Generative AI in zero-day attack detection.

## 3.2 Analysis of Academic and Technical Sources

To evaluate the content generated by Claude 3, a systematic comparison was conducted against a broad range of academic and technical sources referenced in the first part of this study. The objective was to assess both the accuracy of the information provided and its alignment with established scientific knowledge in the field of cybersecurity.

The model's description of zero-day attacks and its emphasis on their critical nature were found to be fully consistent with foundational academic literature, such as the work of Bilge and Dumitras (2012). In particular, Claude 3 accurately highlighted the absence of available patches for such vulnerabilities and their high value within the cybercrime ecosystem—points that are well supported by existing research.

Regarding Generative AI technologies, the model demonstrated a strong understanding of core concepts. Its presentation of Generative Adversarial Networks (GANs) was aligned with the seminal work of Goodfellow et al. (2014), while its discussion of Variational Autoencoders (VAEs) for anomaly detection was consistent with the approach described by An and Cho (2015). These observations indicate a satisfactory level of technical correctness in the model's output.

Furthermore, the model's reference to the role of Large Language Models (LLMs) in cybersecurity aligns with more recent studies, such as Brown et al. (2023), which emphasize their potential in analyzing complex data and supporting threat prediction.

Of particular interest was the inclusion of practical examples, such as the ExplainDet system (Li et al., 2022) and Microsoft's CyberDefender platform (Brown et al., 2024). These references enriched the analysis by introducing a real-world perspective and demonstrating the applicability of the discussed technologies in operational environments.

However, certain limitations were identified. In particular, some claims—especially those related to detection performance—were presented with a degree of vagueness. The absence of detailed contextual information, such as experimental conditions or dataset specifications, limits the reliability of these statements and necessitates further validation through primary sources.

Overall, the content generated by Claude 3 can be considered scientifically valid and largely consistent with existing knowledge in the field. Nevertheless, as is typical when using Generative AI tools, its outputs should be critically evaluated and verified by human experts, especially in high-stakes domains such as cybersecurity.

## 3.3 Comparative Evaluation and Remarks

The comparison between the outputs of ChatGPT and Claude 3 revealed notable differences in both analytical approach and depth of content. ChatGPT's response is characterized by strong academic rigor, featuring a well-structured presentation, precise technical terminology, and a clear emphasis on theoretical foundations. Its analysis is primarily oriented toward conceptual clarity and the systematic explanation of underlying technological principles.

In contrast, Claude 3 adopts a more application-oriented perspective, placing greater emphasis on real-world use cases and practical scenarios. The inclusion of concrete examples—particularly in relation to GANs and Large Language Models (LLMs) in cybersecurity contexts—enhances the practical relevance of its output and improves accessibility, even for non-specialist audiences.

However, Claude 3's responses exhibit a slightly lower level of formal academic documentation. In particular, there is less explicit use of bibliographic references and limited discussion of experimental conditions or empirical validation. As a result, its output tends to be more narrative in style, with occasional reductions in precision compared to the more structured approach of ChatGPT.

Overall, the evaluation suggests that the two models are complementary rather than competitive. ChatGPT excels in producing theoretically grounded and academically structured content, whereas Claude 3 provides a broader perspective by integrating practical insights and real-world applicability. The combined use of both models can therefore offer a more comprehensive understanding of complex topics such as zero-day attack detection using Generative AI.

*Table 1: Comparative Evaluation of ChatGPT and Claude 3*

| **Criterion** | **ChatGPT** | **Claude 3** | **Comment** |
|---|---|---|---|
| Technical Depth | High | Medium-High | ChatGPT provides more structured and technically rigorous explanations |
| Practical Examples | Medium | High | Claude includes more real-world scenarios and applied context |
| Clarity & Readability | High | High | Both models produce clear and understandable content |

| | | | |
|---|---|---|---|
| Academic Referencing | Medium-High | Medium | ChatGPT aligns more with formal academic tone and structure |
| Level of Detail | High | Medium-High | ChatGPT focuses more on detailed theoretical explanation |
| Risk of Generalization | Medium | Medium-High | Claude tends to be more narrative and occasionally less precise |

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## 3.4 Limitations of this Study

Despite the effort to provide a comprehensive and well-structured analysis, the present study is subject to several limitations that should be carefully considered when interpreting its findings.

First, the adopted approach is primarily theoretical, relying on a literature review and the analysis of content generated by Generative AI models. No implementation of a fully functional zero-day attack detection system was undertaken, nor was any empirical evaluation conducted using real-world cybersecurity datasets.

Furthermore, the comparative evaluation between ChatGPT and Claude 3 is based mainly on qualitative criteria, focusing on the content, structure, and depth of the generated responses. Quantitative performance metrics—such as precision, recall, and false positive rates—were not assessed, as their evaluation would require a controlled experimental setup.

Another limitation concerns the reliance on Generative AI tools themselves. Such systems may occasionally produce incomplete, imprecise, or insufficiently documented outputs, making human oversight and verification through reliable sources essential.

In addition, the study does not extensively address practical implementation challenges in real-world environments, including issues related to computational cost, scalability, and integration with existing cybersecurity infrastructures.

These limitations indicate that the conclusions of this study should be interpreted with caution and viewed as indicative rather than definitive. Nevertheless, they provide a useful foundation for further research and more comprehensive experimental investigation.

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## 3.5 Conclusions

The comparative evaluation of ChatGPT and Claude 3 demonstrates that Generative AI tools can serve as highly valuable instruments for the analysis and understanding of complex cybersecurity challenges, such as zero-day attack detection. Despite the significant progress achieved in terms of accuracy and functionality, their use must be accompanied by a clear awareness of their limitations and supported by systematic validation.

ChatGPT proved particularly effective in delivering structured, theoretically grounded analysis and in clearly presenting technical concepts. In contrast, Claude 3 contributed a stronger emphasis on practical application, offering richer real-world examples and illustrating how theoretical concepts can be applied in operational contexts. However, in both cases, human oversight, critical thinking, and verification against reliable academic sources remain essential.

Overall, the findings suggest that the effective use of Generative AI in cybersecurity depends on the balanced integration of technological capabilities and rigorous academic methodology. While these tools hold considerable promise for supporting critical domains such as cyber-threat detection, their future impact will depend on the development of more transparent, explainable, and reliable systems.

# Appendix A: Prompts Used

## A.1 Prompts for ChatGPT (OpenAI)

**Description:** ChatGPT was utilized to generate the primary content of Part A of the study (*"Theoretical Analysis and Current Applications"*).

**Prompt given:** "Write an analytical paper on the topic: Generative AI for Zero-Day Attack Detection. The paper should include:

→The definition and characteristics of zero-day attacks.
→An explanation of the basic operating principles of Generative AI.
→ How Generative AI models can detect zero-day attacks.
→ Examples of applications, including GANs, VAEs, LLMs, XDR, and Explainable AI (XAI)
→ A discussion of the advantages and limitations of this approach"

## A.2 Prompts for Claude 3 (Anthropic)

**Description:** Claude 3 was utilized to generate content for Part B of the study (*"Comparative Evaluation of AI"*) and to support the cross-validation of the analysis.
**Prompt given**: "Write an academic analysis on the topic: Generative AI for Zero-Day Attack Detection. The analysis should include:

→ A definition and discussion of the risks associated with zero-day attacks
→ An explanation of key concepts and architectures of Generative AI (e.g., GANs, VAEs, Transformers)
→ Cybersecurity use cases, including real-world examples.
→ An evaluation of the advantages, limitations, and future outlook of Generative AI in zero-day attack detection

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